$$E_n = -\frac{m_e e^4}{32\pi^2 \varepsilon_0^2 \hbar^2 n^2}. \qquad (10)$$

7. (10) $n = 1$,

$$E_1 = -\frac{m_e e^4}{32\pi^2 \varepsilon_0^2 \hbar^2} = -2.24 \times 10^{-18} \text{ Дж}. \qquad (11)$$

8. (9) :

$$\frac{dE}{dt} = -P. \qquad (12)$$

(9) (12) :

$$r(t)^2 \frac{dr(t)}{dt} = -\frac{e^4}{12\pi^2 \varepsilon_0^2 m_e^2 c^3}. \qquad (13)$$

(13) $r(0) = r_1$:

$$r(t) = \sqrt[3]{r_1^3 - \frac{e^4}{4\pi^2 \varepsilon_0^2 m_e^2 c^3} t}. \qquad (14)$$

9. τ_1 : $r(t) = 0$. (14)

$$\tau_1 = \frac{4\pi^2 m_e^2 \varepsilon_0^2 c^3 r_1^3}{e^4} = \frac{256\pi^5 \varepsilon_0^5 c^3 \hbar^5}{m_e e^{10}} = 1.44 \times 10^{-11} \text{ с}. \qquad (15)$$

10. dt : $d\varphi$:

$$d\varphi = \frac{\text{V}}{r} dt. \qquad (16)$$

(5) dt (13) :

$$d\varphi = -\frac{6c^3 (\pi m_e \varepsilon_0)^{3/2}}{e^3} \sqrt{r}\, dr. \qquad (17)$$

r_1 :

$$\varphi = \int_{r_1}^{0} d\varphi = \int_{0}^{r_1} \frac{6c^3 (\pi m_e \varepsilon_0)^{3/2}}{e^3} \sqrt{r}\, dr = \frac{4c^3 (\pi m_e \varepsilon_0 r_1)^{3/2}}{e^3} = \frac{32\pi^3 \varepsilon_0^3 c^3 \hbar^3}{e^6}. \qquad (18)$$

:

$$N = \frac{\varphi}{2\pi} = \frac{16\pi^2 \varepsilon_0^3 c^3 \hbar^3}{e^6} = 1.96 \times 10^5. \qquad (19)$$

채점 기준

№	평가 항목	배점	
1	(1)	0.5	
	(2)	0.5	
	(3)	0.5	**2.0**
	t_1	0.5	
2	(4)	0.5	**1,0**
	(5)	0.5	
3	(6)	0.5	**0.5**
4	(7)	0.5	**0.5**
5	r_1	0.5	**0.5**